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VARUNA AI

- Problem Statement ID – SIH260006
- Problem Statement Title- Explainable AI Model to Predict High-Impact Rain Events Using Satellite Data
- Theme- Space Technology
- PS Category- Software
- Team Name- ALL IN

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VARUNA AI

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- **Problem:** Cloudbursts and extreme rain events cause floods and landslides across India; current forecasts are hard to trust or act on quickly
- **VARUNA AI:** an Explainable AI system that predicts high-impact rainfall using ISRO INSAT satellite data and weather data
- Gives a **1–72 hour** rainfall-risk forecast, classified **Low to Extreme** with a confidence score
- Explains each prediction using **SHAP and Grad-CAM** on the satellite image
- Shows risk and early-warning alerts on an interactive **India risk map** for disaster-management teams
- **Core innovation:** it doesn't just predict heavy rain — it **explains why** the AI believes the risk is high

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TECHNICAL APPROACH

Explainable AI based Extreme Rainfall Intelligence & Early Warning System

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1. DATA SOURCES

Satellite Data (NASA GIBS)

- Precipitation
- Cloud Cover
- Humidity
- Temperature
- Wind Speed

Weather Data (APIs)

- Forecast Data
- Historical Data
- Real-time Data

2. DATA INGESTION & PROCESSING

Data Ingestion
(FastAPI Scheduler)

Raw Data Storage
PostgreSQL / PostGIS
(Spatial + Tabular)

Data Processing
Cleaning, Resampling,
Feature Engineering

Prepared Data
Tabular + Raster
(Analysis Ready)

3. AI & ANALYTICS LAYER

AI Models

- Tabular Models (XGBoost, Random Forest)
- Vision Models (CNN for Satellite Images)
- Hybrid Models

Prediction Engine

- Rainfall Intensity Prediction
- Risk Classification (Low / Moderate / High)

Alert Engine

- Threshold Based Alerts
- Early Warning Generation

4. EXPLAINABILITY LAYER

SHAP Analysis

- Feature Importance
- Global & Local Interpretations

Grad-CAM

- Visual Explanations for Satellite Images
- Heatmaps

Insights & Reports

- Model Insights
- Explanation Reports
- Downloadable PDFs

5. APPLICATION LAYER

Web Dashboard (React)

- Interactive Maps
- Rainfall Forecasts
- Trends & Analytics
- Alerts & Notifications

Notifications

- Email Alerts
- SMS / Push Alerts

User Roles

- Admin
- Analyst
- Public User

TECHNOLOGY STACK

Python	(Backend Development)
FastAPI	(API & Scheduler)
PostgreSQL + PostGIS	(Spatial DB)
PyTorch	(Deep Learning (Models)
scikit-learn	(ML Models)
SHAP	(Explainable AI)
React	(Frontend)
Docker	(Containerization)
Leaflet	(Maps)

INFRASTRUCTURE & DEPLOYMENT

AWS Cloud	(EC2, S3, IAM)
Docker	(Containerization)
PostgreSQL + PostGIS	(Geospatial DB)
FastAPI	(Backend API)
React	(Frontend)
GitHub	(Code Repository)

Legend: → Data Flow - - - → Deployment / Integration

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FEASIBILITY AND VIABILITY

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Feasibility

- Required satellite and weather data sources are available
- Models can be developed and validated in stages
- Modular architecture allows independent testing

Challenges

- Missing or delayed data
- False alarms
- Model interpretability
- Operational decision-making

Mitigation

- Data validation and fallback sources
- Historical validation and threshold tuning
- SHAP and Grad-CAM explanations
- Human-in-the-loop decision support

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IMPACT AND BENEFITS

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Target Users

- Disaster-management authorities
- Emergency-response teams
- Weather and research teams

Benefits

- Earlier identification of high-impact rainfall risk
- Region-level risk information for faster decisions
- Better understanding of AI predictions through explanations
- Human-readable, easy-to-act-on warnings
- Prediction and alert history for auditing

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RESEARCH AND REFERENCES

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- **MOSDAC / ISRO Satellite Data:** https://www.mosdac.gov.in/
- **NASA Earthdata:** https://www.earthdata.nasa.gov/
- **SHAP Documentation:** https://shap.readthedocs.io/en/latest/
- **Grad-CAM Paper (ICCV 2017):**

  https://openaccess.thecvf.com/content_iccv_2017/html/Selvaraju_Grad-

  CAM_Visual_Explanations_ICCV_2017_paper.html
- **Project GitHub Repository:** https://github.com/Unnathihegde/isrosih